A Complete Constraint Count for the Regular Higher-Derivative S004 Complex Scalar

Richard J. Reyes

October 26, 2026

Abstract

For the specified nonzero-field, fixed-flat-background sector of the corrected S004 complex-scalar action, the Hessian with respect to the two real accelerations is $-2\kappa F I_2$, with nonzero determinant $4\kappa^2 F^2$. An explicit first-order Dirac reduction produces four second-class primary constraints and no secondary or first-class scalar constraints. Starting from twelve first-order canonical variables, the reduced phase space therefore has eight real dimensions per spatial point, corresponding to four real configuration degrees of freedom. Two degrees are additional to those of an ordinary complex scalar. The conserved global $U(1)$ charge is not a local gauge constraint. The result applies only to the specified regular fixed-background sector; it does not analyze an unspecified metric or gauge completion or the degenerate vacuum.

1 Hypotheses and action

Use the rescaled time coordinate $\tau = c_0 t$ and the flat metric of signature $(-,+,\dots,+)$. Write

$$\psi = X_1 + iX_2, \qquad \mathbf{X} = (X_1, X_2), \qquad \rho = |\psi|^2 = |\mathbf{X}|^2. \tag{1}$$

Define

$$D(\rho) = \rho + \varepsilon^2 e^{-2\alpha\rho}, \qquad F(\rho) = \frac{\rho}{D(\rho)^2}. \tag{2}$$

Assume

$$\varepsilon > 0, \qquad \kappa > 0, \qquad \theta > 0, \qquad \gamma^2 < 4\kappa\theta, \tag{3}$$

and restrict the canonical analysis to a smooth field-space patch $\rho > 0$ (bounded away from zero when uniform regularity is required). Then $F(\rho) > 0$. The scalar background metric and foliation are fixed; no independent metric, connection, or gauge field is included.

The original S004 scalar density is

$$\mathcal{L}_{\text{orig}} = \partial_\mu \bar{\psi}\, \partial^\mu \psi - V(\rho) + F(\rho)\left[\kappa|\Box\psi|^2 + \theta|\Delta\psi|^2 + \gamma\,\mathrm{Re}(\Box\psi\,\overline{\Delta\psi})\right]. \tag{4}$$

We use $\mathcal{L}_{\text{phys}} = -\mathcal{L}_{\text{orig}}$, which orients the ordinary two-derivative kinetic term positively. An overall sign reversal does not alter the constraint count.

Put

$$\mathbf{V} = \partial_\tau \mathbf{X}, \quad \mathbf{a} = \partial_\tau^2 \mathbf{X}, \quad \mathbf{b} = \Delta\mathbf{X}, \quad A = 2\kappa + \gamma, \quad C = \kappa + \theta + \gamma. \tag{5}$$

Because $\Box\psi = -\mathbf{a} + \mathbf{b}$ in real components,

$$|\Box\psi|^2 = |\mathbf{a}|^2 - 2\mathbf{a}\cdot\mathbf{b} + |\mathbf{b}|^2, \tag{6}$$

$$\mathrm{Re}(\Box\psi\,\overline{\Delta\psi}) = -\mathbf{a}\cdot\mathbf{b} + |\mathbf{b}|^2. \tag{7}$$

It follows that

$$\boxed{\mathcal{L}_{\rm phys} = |\mathbf{V}|^2 - |\nabla \mathbf{X}|^2 + V(\rho) - F(\rho)\left(\kappa|\mathbf{a}|^2 - A\mathbf{a}\cdot\mathbf{b} + C|\mathbf{b}|^2\right).} \tag{8}$$

We assume boundary conditions sufficient to justify the integrations by parts and the Hamiltonian manipulations below (e.g., periodic spatial data or suitably decaying variations with vanishing boundary flux).

2 The acceleration Legendre map

Lemma 1 (Regular acceleration Hessian)**.** *On the patch $\rho > 0$, the real two-by-two acceleration Hessian obeys*

$$H_{\rm acc} = \left(\frac{\partial^2 \mathcal{L}_{\rm phys}}{\partial a_i \partial a_j}\right)_{i,j=1}^{2} = -2\kappa F(\rho) I_2, \qquad \det H_{\rm acc} = 4\kappa^2 F(\rho)^2 > 0. \tag{9}$$

Thus the higher-time Legendre map is pointwise invertible.

Proof. Differentiating equation (8) with $\mathbf{X}$, $\mathbf{V}$, and $\mathbf{b}$ fixed gives

$$\frac{\partial \mathcal{L}_{\rm phys}}{\partial a_i} = -F(2\kappa a_i - A b_i). \tag{10}$$

Since F depends on $\mathbf{X}$, not on $\mathbf{a}$, one more derivative gives $\partial^2 \mathcal{L}_{\rm phys}/\partial a_i \partial a_j = -2\kappa F \delta_{ij}$. The determinant is positive under (3) and $\rho > 0$. □

3 First-order Dirac reduction

Introduce $\mathbf{V}$ as an independent configuration variable and a real two-component multiplier $\boldsymbol{\lambda}$ imposing $\dot{\mathbf{X}} = \mathbf{V}$:

$$\mathcal{L}_1 = \mathcal{L}_{\rm phys}(\mathbf{X}, \mathbf{V}, \dot{\mathbf{V}}, \Delta\mathbf{X}) + \boldsymbol{\lambda}\cdot(\dot{\mathbf{X}} - \mathbf{V}). \tag{11}$$

The six real first-order configuration coordinates are $(\mathbf{X}, \mathbf{V}, \boldsymbol{\lambda})$, with conjugate momenta $(\mathbf{p}_X, \mathbf{p}_V, \mathbf{p}_\lambda)$. The momentum equations are

$$\mathbf{p}_X = \boldsymbol{\lambda}, \qquad \mathbf{p}_\lambda = 0, \qquad \mathbf{p}_V = -F(2\kappa\dot{\mathbf{V}} - A\mathbf{b}). \tag{12}$$

By Lemma 1, the final relation is solvable:

$$\dot{\mathbf{V}} = \frac{A}{2\kappa}\mathbf{b} - \frac{\mathbf{p}_V}{2\kappa F}. \tag{13}$$

Hence no primary constraint arises from $\mathbf{p}_V$. The multiplier sector provides exactly four primary constraints,

$$\Phi_i = p_{X_i} - \lambda_i \approx 0, \qquad \Xi_i = p_{\lambda_i} \approx 0, \qquad i = 1, 2. \tag{14}$$

Lemma 2 (All four primary constraints are second class)**.** *The Dirac matrix of the four constraints in (14) has rank four. In particular, no nonzero first-class linear combination of these primary constraints exists.*

Proof. The equal-time canonical brackets are

$$\{X_i(\mathbf{y}), p_{X_j}(\mathbf{z})\} = \{V_i(\mathbf{y}), p_{V_j}(\mathbf{z})\} = \{\lambda_i(\mathbf{y}), p_{\lambda_j}(\mathbf{z})\} = \delta_{ij}\delta(\mathbf{y}-\mathbf{z}). \tag{15}$$

The nonzero brackets among the constraints are

$$\{\Phi_i(\mathbf{y}), \Xi_j(\mathbf{z})\} = -\delta_{ij}\delta(\mathbf{y}-\mathbf{z}). \tag{16}$$

Suppressing the common spatial delta distribution, the matrix in the order $(\Phi_1, \Phi_2, \Xi_1, \Xi_2)$ is

$$\mathcal{C} = \begin{pmatrix} 0 & -I_2 \\ I_2 & 0 \end{pmatrix}, \qquad \det \mathcal{C} = 1. \tag{17}$$

Thus all four constraints are second class. □

4 Preservation and termination of the Dirac algorithm

The partial Legendre transform in the acceleration variable, using (13), gives

$$\boxed{\mathcal{H}_0 = -|\mathbf{V}|^2 + |\nabla \mathbf{X}|^2 - V(\rho) - \frac{|\mathbf{p}_V|^2}{4\kappa F} + \frac{A}{2\kappa}\mathbf{p}_V \cdot \mathbf{b} + B_\perp F |\mathbf{b}|^2,} \tag{18}$$

where

$$B_\perp = C - \frac{A^2}{4\kappa} = \theta - \frac{\gamma^2}{4\kappa} > 0. \tag{19}$$

The total Hamiltonian takes the form

$$H_T = \int \left[\mathcal{H}_0 + \boldsymbol{\lambda}\cdot\mathbf{V} + \mathbf{u}\cdot\boldsymbol{\Phi} + \mathbf{w}\cdot\boldsymbol{\Xi}\right] \mathrm{d}^d\mathbf{y}. \tag{20}$$

For each $i = 1, 2$, preservation of Ξ_i yields

$$\dot{\Xi}_i = \{\Xi_i, H_T\} = u_i - V_i \approx 0, \qquad \Longrightarrow \qquad u_i = V_i. \tag{21}$$

Preservation of Φ_i yields

$$\dot{\Phi}_i = \{\Phi_i, H_T\} = -\frac{\delta H_0}{\delta X_i} - w_i \approx 0, \qquad \Longrightarrow \qquad w_i = -\frac{\delta H_0}{\delta X_i}, \tag{22}$$

with $H_0 = \int \mathcal{H}_0 \mathrm{d}^d\mathbf{y}$. The functional derivative includes all spatial-derivative terms. Both preservation conditions fix Lagrange multipliers; neither imposes a new condition on $(\mathbf{X}, \mathbf{V}, \mathbf{p}_X, \mathbf{p}_V)$. There are therefore no secondary constraints, and the Dirac algorithm terminates.

5 Degree-of-freedom theorem

Theorem 1 (Four regular real configuration degrees of freedom)**.** *Under the hypotheses in Section 1, the fixed-background corrected S004 complex-scalar sector has four local real configuration degrees of freedom per spatial point. Two are additional higher-time degrees relative to an ordinary complex scalar. There are four second-class scalar constraints, no first-class scalar constraints, and no secondary scalar constraints.*

Proof. The auxiliary first-order formulation contains six real configuration coordinates, hence twelve real canonical phase-space dimensions per spatial point. The invertible acceleration Hessian introduces no extra primary constraint; equations (14)–(17) give four second-class constraints. The preservation equations above show that no secondary constraints arise and that there are no first-class scalar constraints. Dirac's counting formula therefore gives

$$N_{\mathrm{conf}} = \frac{N_{\mathrm{phase}} - 2N_{\mathrm{first}} - N_{\mathrm{second}}}{2} = \frac{12 - 2(0) - 4}{2} = 4. \tag{23}$$

Equivalently, eliminating $(\boldsymbol{\lambda}, \mathbf{p}_\lambda)$ leaves four unconstrained canonical pairs,

$$(X_1, p_{X_1}), \quad (X_2, p_{X_2}), \quad (V_1, p_{V_1}), \quad (V_2, p_{V_2}), \tag{24}$$

so the reduced phase space has eight real dimensions per spatial point. The two $\mathbf{V}$ components constitute two additional local higher-time configuration degrees of freedom. □

6 Canonical energy and global phase charge

On the reduced phase space, the exact canonical Hamiltonian is

$$\boxed{H_{\mathrm{phys}} = \int \Big[\mathbf{p}_X \cdot \mathbf{V} - |\mathbf{V}|^2 + |\nabla \mathbf{X}|^2 - V(\rho) \\ - \frac{|\mathbf{p}_V|^2}{4\kappa F} + \frac{A}{2\kappa} \mathbf{p}_V \cdot \Delta \mathbf{X} + B_\perp F |\Delta \mathbf{X}|^2 \Big] \mathrm{d}^d \mathbf{y}.} \tag{25}$$

The term $\mathbf{p}_X \cdot \mathbf{V}$ is linear in an unconstrained momentum; moreover, $-|\mathbf{p}_V|^2/(4\kappa F)$ is negative for nonzero $\mathbf{p}_V$. These are explicit manifestations of the unbounded-energy obstruction in this nondegenerate higher-time sector. This energy observation is separate from, and not required for, the counting theorem.

Let $J = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. The constant phase transformation $\psi \mapsto e^{i\eta}\psi$ acts as $\delta \mathbf{X} = \eta J \mathbf{X}$ and $\delta \mathbf{V} = \eta J \mathbf{V}$. Its canonical generator is the conserved global charge

$$\boxed{\mathcal{Q} = \int (\mathbf{p}_X \cdot J\mathbf{X} + \mathbf{p}_V \cdot J\mathbf{V}) \, \mathrm{d}^d \mathbf{y}.} \tag{26}$$

Conservation presupposes sufficient solution regularity, integrability, and vanishing boundary flux. Importantly, a *constant* $U(1)$ parameter is not a free spacetime-dependent gauge parameter. The charge (26) is not a local first-class constraint, and fixing its value does not eliminate the two extra local degrees of freedom.

7 Limitations

Remark 1 (Degenerate vacuum)**.** *At $\psi = 0$ one has $F(0) = 0$, so the acceleration Hessian loses rank. The regular-sector Legendre map is then inapplicable. A separate singular constraint analysis is required at that point; it cannot be used to eliminate modes throughout neighboring regular nonzero-field patches.*

Remark 2 (Different theories and completions)**.** *The theorem concerns the literal corrected S004 scalar sector with fixed flat metric and fixed foliation. It does not supply or count constraints of an unspecified dynamical-metric, connection, or gauge completion. The $\kappa = 0$ case likewise requires a separate analysis. For global infinite-domain solutions, one must additionally control the behavior of F and appropriate canonical function spaces; the count above is a local regular-sector count.*

Conclusion

For $\kappa > 0$ and $F(\rho) > 0$,

$$\boxed{H_{\mathrm{acc}} = -2\kappa F I_2, \quad \det H_{\mathrm{acc}} = 4\kappa^2 F^2 > 0, \quad N_{\mathrm{phase,red}} = 12 - 4 = 8, \quad N_{\mathrm{conf}} = 4.} \tag{27}$$

The two extra real higher-time configuration degrees of freedom survive all scalar constraints of the specified fixed-background system. A global $U(1)$ charge does not remove them.

References

[1] R. J. Reyes, "S004 physical-health audit: constraints, ghost and ultraviolet obstruction," October 20, 2026, https://github.com/rickyjreyes/geometry_of_resonance/blob/main/docs/foundational/GHOST_AND_CONSTRAINT_ANALYSIS.md.

[2] R. P. Woodard, "The Theorem of Ostrogradsky," *Scholarpedia* **10**(8), 32243 (2015), https://arxiv.org/abs/1506.02210.